IBM Z DATATHON 2026 · TEAM CHINESE CAN FLY

Real-Time Transfer Scam Guard

Stopping scam-driven transfers inside the bank's authorisation path — before the money leaves.

Track: Real-Time AI for Critical Decisions · Core principle: data analysed immediately → AI influences decisions while it matters

APPROVE
release

CONFIRM
pause + alert the family

BLOCK
hold + audit record

Runs on IBM LinuxONE (s390x) · Telum on-chip AI · pervasive encryption · container native

THE PROBLEM

Older people send the money themselves

Victim tricked
phone scam, hours of pressure

›

Transfer authorised
valid PIN, valid device

›

Funds cleared
irreversible

›

Bank / family notice
too late

The only effective intervention point is the few seconds BEFORE authorisation.

It looks normal to the bank

Valid credentials, valid device, correct PIN. Nothing in the payment itself is anomalous.

Existing systems target card fraud

They detect misuse after the fact. They are not built for a customer being talked into paying right now.

The family is never involved

The people who could actually stop it — the adult children — have no way in.

THE SOLUTION

Three outcomes, decided in the authorisation path

APPROVE

Release immediately

Risk below 0.55.
No friction added to normal payments.

CONFIRM

Pause · message the holder · alert the family

Risk 0.55–0.95.
The holder gets plain language.
If the amount exceeds the guardian limit, a linked relative is alerted too.

BLOCK

Hold and record

Risk above 0.95.
Transfer held; an auditable record is written with reason codes.

The part nobody else does

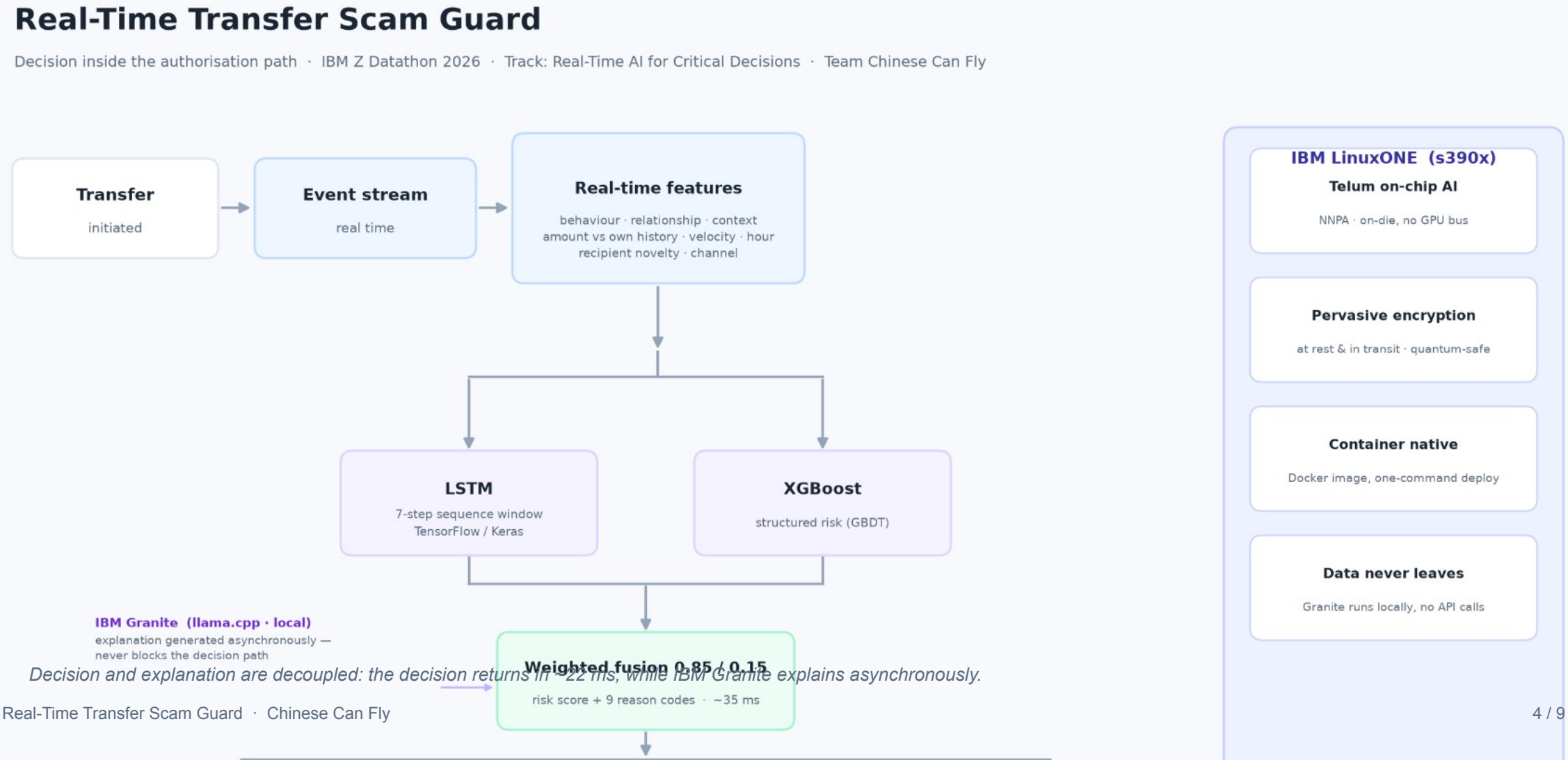
CONFIRM is not just a block. It recruits the one person who can actually stop the scam: the family.

Every decision is auditable

Nine reason codes (R01–R09) travel with each decision, so a case can be reviewed and the model audited later.

ARCHITECTURE

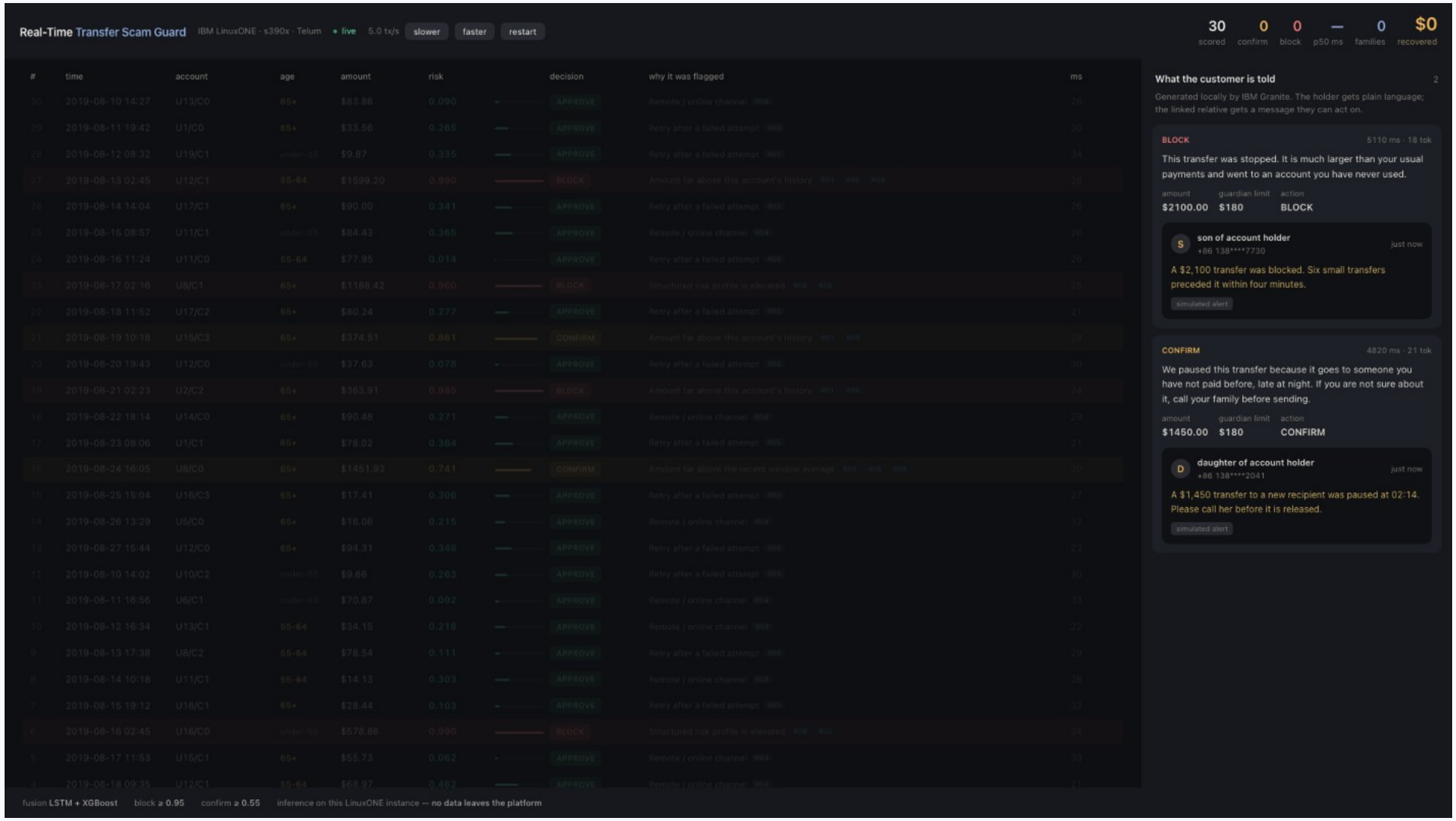
Two models, one decision, ~22 ms



Decision and explanation are decoupled: the decision returns in ~22 ms, while IBM Granite explains asynchronously.

THE CONSOLE

This is the screen, running on LinuxONE



Every row is a live decision. Right lane: the plain-language message the holder sees, and the alert the linked relative receives.

WHY IBM Z / LINUXONE

Not an x86 app ported over

Telum on-chip AI

Both models and the Granite LLM run on the same processor. The accelerator sits on-die, not behind a GPU bus — which is why the decision stays fast even while the LLM generates.

Pervasive encryption

Data encrypted at rest and in transit by the platform, quantum-safe. Scam detection must read sensitive transfer and identity context, so this is a prerequisite.

Data never leaves

Granite is served locally by llama.cpp with the zDNN/NNPA backend. No external API calls. No customer data leaves the enterprise perimeter.

Container native

Shipped as a Docker image built on IBM's s390x AI/ML base image. One command to deploy; health-checked and restart-safe.

RESULTS

Measured on the LinuxONE instance

65.2 %
fraud recall

96.7 %
precision on stops

22 ms
median decision

$18,418
amount recovered

Offline model quality (held-out test, 0.11 % positives)

Model	PR-AUC	ROC-AUC	Best F1
LSTM only	0.3829	0.9293	0.478
XGBoost only	0.0765	0.9411	0.161
Fusion (0.85 / 0.15)	**0.4277**	**0.9654**	**0.506**

Why fusion

Every single model leaves money on the table:

- XGBoost alone catches few frauds at usable precision
- LSTM alone misses structured signals such as amount versus the account's own history
- The fusion beats both on PR-AUC and on F1, which is exactly why two frameworks are used rather than one.

587 replayed transfers (single reproducible pass) · 3 vCPU + 1 AI accelerator · 587-transfer curated stream including 271 labelled frauds

WHO BENEFITS

Four audiences, in order of directness

1

Older adults targeted by scams

They act alone and often cannot stop themselves in time. The system pauses the transfer and offers a plain way to say "I am not sure" instead of losing their savings.

2

Adult children living away

They cannot watch every transfer. They set a guardian limit and get an actionable alert, turning distant worry into a decision they can make from their phone.

3

Banks and payment providers

Fewer scam losses and complaints, plus a defensible audit record for regulators. The engine sits in the authorisation path, so it cuts loss without slowing genuine payments.

4

Anti-fraud teams and regulators

Every decision carries reason codes. Cases can be reviewed later and the model audited over time — a requirement in most regulated markets.

SCOPE

What this prototype is — and is not

What it does

- Streams transfers and scores each one in ~22 ms
- Three-way decision with reason codes
- Plain-language message for the holder
- Guardian alert when the limit is exceeded
- Fully containerised, runs on LinuxONE
- Granite explanations generated locally, no API calls

What it does not do (yet)

- No live bank core integration — the stream is historical replay
- Guardian notification is simulated, not a real SMS gateway
- No production load testing at peak volume
- Age and guardian fields are a synthetic scenario overlay
- onnxruntime has no s390x wheel, so ONNX is export-only

Next 30 days: swap the replay for the bank's authorisation feed, connect a real notification provider, and run IEEE-CIS + PaySim as the production corpus with time-drift and fairness checks.